Results

Candidate Outcome

Candidate Outcome I

The generated loop selected exp-mlp-tanh-64 (One-hidden-layer tanh MLP, MLP) after evaluating 4 candidate(s) from a proposed set of 5. The nearest_centroid_baseline (nearest-centroid) baseline reached 82.6% test_accuracy, while the selected candidate reached 89.4%, an absolute change of 6.8%. The selected model has 50890 parameters. The transformer-candidate evaluated flag is true, and the candidate budget exhausted flag is true, which means the ledger records 1 deferred proposal(s) rather than expanding the run automatically.

Candidate Outcome I

The benchmark score is 1. That score is not a model-quality claim by itself; it is a compact grading artifact for the methods contract: metric improvement, budget compliance, offline execution, and selected-candidate recording. Rank-stability diagnostics report that the selected candidate is top ranked in 72.5% of deterministic bootstrap resamples, with runner-up `exp-mlp-relu-32`. The candidate score figure is registered as fig. 6, the confusion matrix as fig. 8, the per-class diagnostic as fig. 9, the learning curves as fig. 10, the complexity diagnostic as fig. 12, the selected-candidate error examples as fig. 13, the candidate lifecycle diagnostic as fig. 14, rank stability as fig. 7, the training-dynamics diagnostic as fig. 11, the final readiness matrix as fig. 25, and the process closure as fig. 26.

Candidate Outcome I

Table 1: Candidate accuracy intervals from `output/data/ml_candidate_intervals.json`; intervals describe the fixed local test split.

Candidate	Status	Correct/test	Accuracy	Wilson 95% interval
baseline	baseline	413/500	82.6%	79.0% to 85.7%
softmax linear	rejected	441/500	88.2%	85.1% to 90.7%
mlp relu 32	rejected	443/500	88.6%	85.5% to 91.1%
tiny patch attention	rejected	152/500	30.4%	26.5% to 34.6%
mlp tanh 64	accepted	447/500	89.4%	86.4% to 91.8%

Candidate Outcome II

Table 2: Candidate rank-stability table from `output/data/ml_candidate_rank_stability.json`; frequencies are deterministic local bootstrap summaries.

Candidate	Observed rank	Top-rank frequency	Mean rank	Accuracy
softmax linear	3	4.2%	2.554	88.2%
mlp relu 32	2	23.3%	2.130	88.6%
tiny patch attention	4	0.0%	4.000	30.4%
mlp tanh 64	1	72.5%	1.316	89.4%

Run-Derived Figures

Run-Derived Figures I

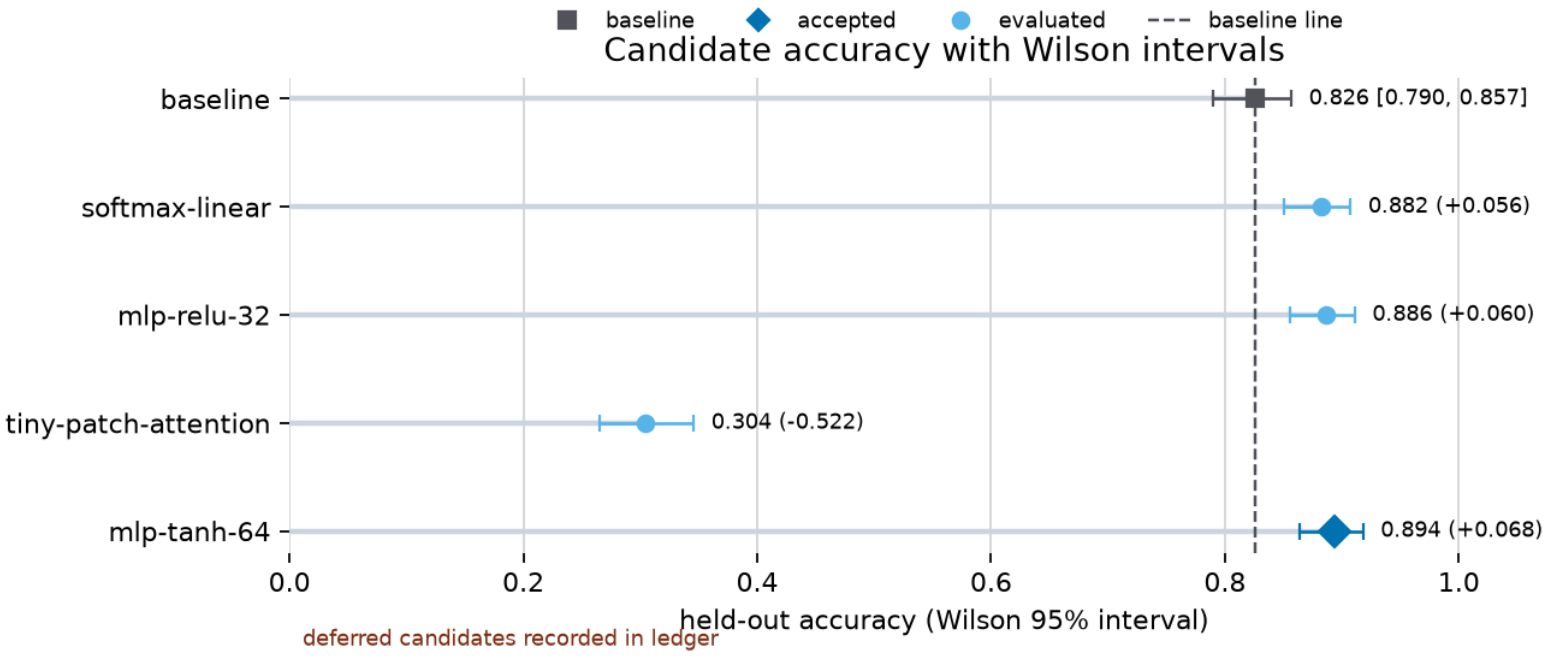
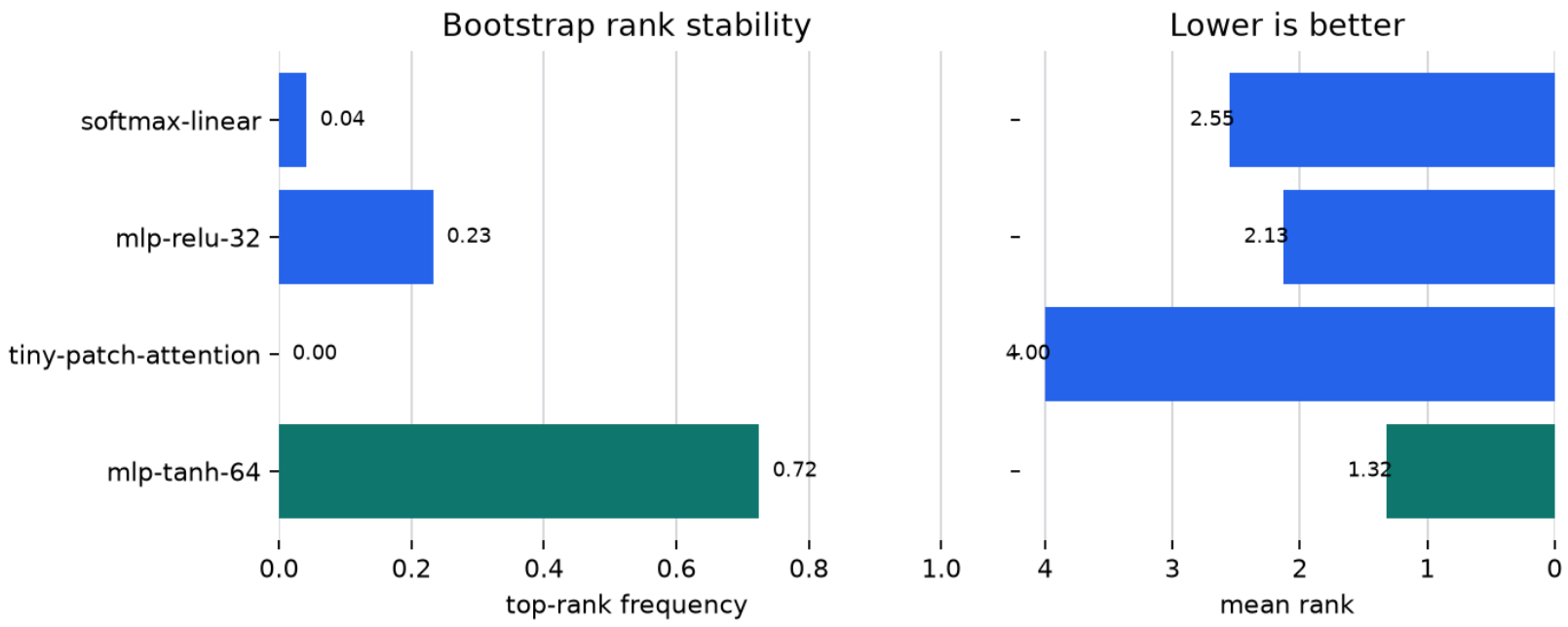


Figure 6: Held-out accuracy with Wilson intervals for the baseline and evaluated candidates from output/data/ml_candidate_intervals.json; accepted candidate exp-mlp-tanh-64 improves accuracy from 82.6% to 89.4% on the fixed subset, with deferred proposals kept in the candidate ledger. Generation method: Lollipop accuracy comparison with Wilson interval error bars and direct labels. Registry metadata records

Run-Derived Figures I

the generation method, source artifact, and claim boundary for validation.

Run-Derived Figures I



accepted top-rank frequency 0.72; runner-up mlp-relu-32

Figure 7: Rank-stability summary for exp-mlp-tanh-64 from output/data/ml_candidate_rank_stability.json; deterministic local bootstrap resampling shows how often each evaluated candidate ranks first under the fixed test split. Generation method: Bootstrap top-rank frequency and mean-rank comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Run-Derived Figures I

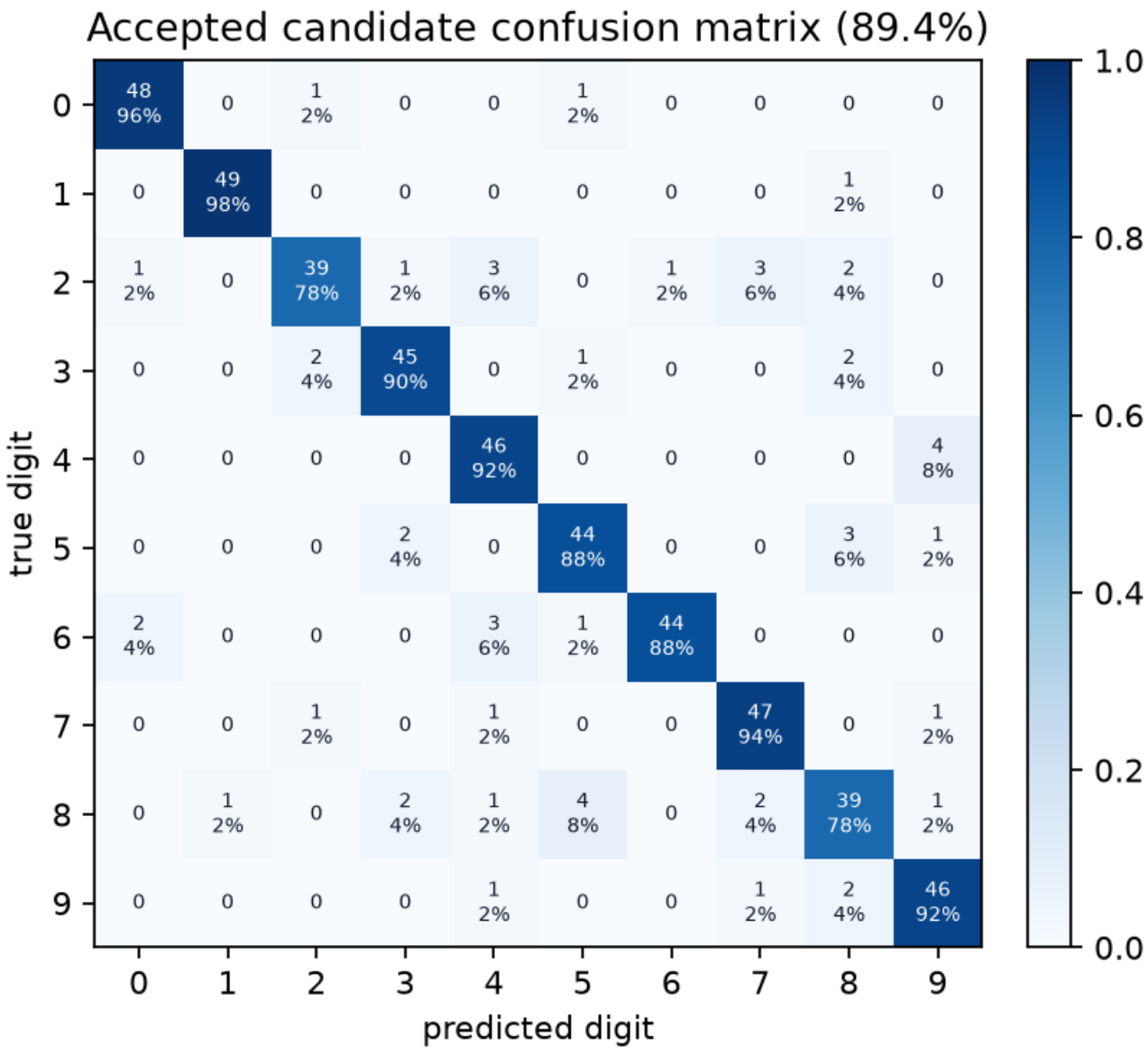


Figure 8: Accepted-candidate confusion matrix for exp-mlp-tanh-64 on the fixed MNIST handwritten digit database test split, sourced from output/data/ml_confusion_matrix.csv; it diagnoses the selected run, not general full-dataset performance. Generation method: Row-normalized heatmap with cell counts and row percentages. Registry metadata records the generation method, source artifact, and claim boundary for

Run-Derived Figures I

validation.

Run-Derived Figures I

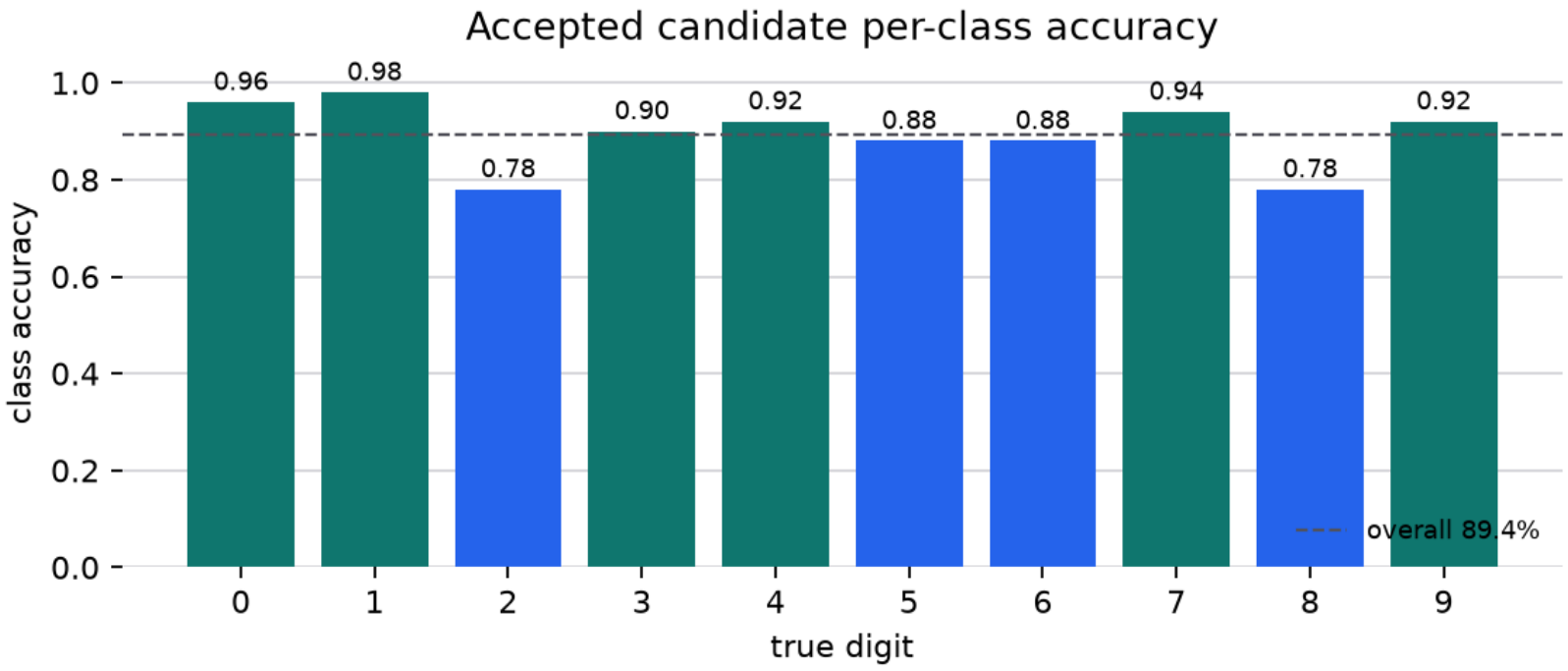


Figure 9: Per-class accuracy for exp-mlp-tanh-64, computed from output/data/ml_confusion_matrix.csv; variation across digits is a run diagnostic for the fixed local test split. Generation method: Per-class accuracy bars computed from the confusion matrix diagonal. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics

Training And Error Diagnostics I

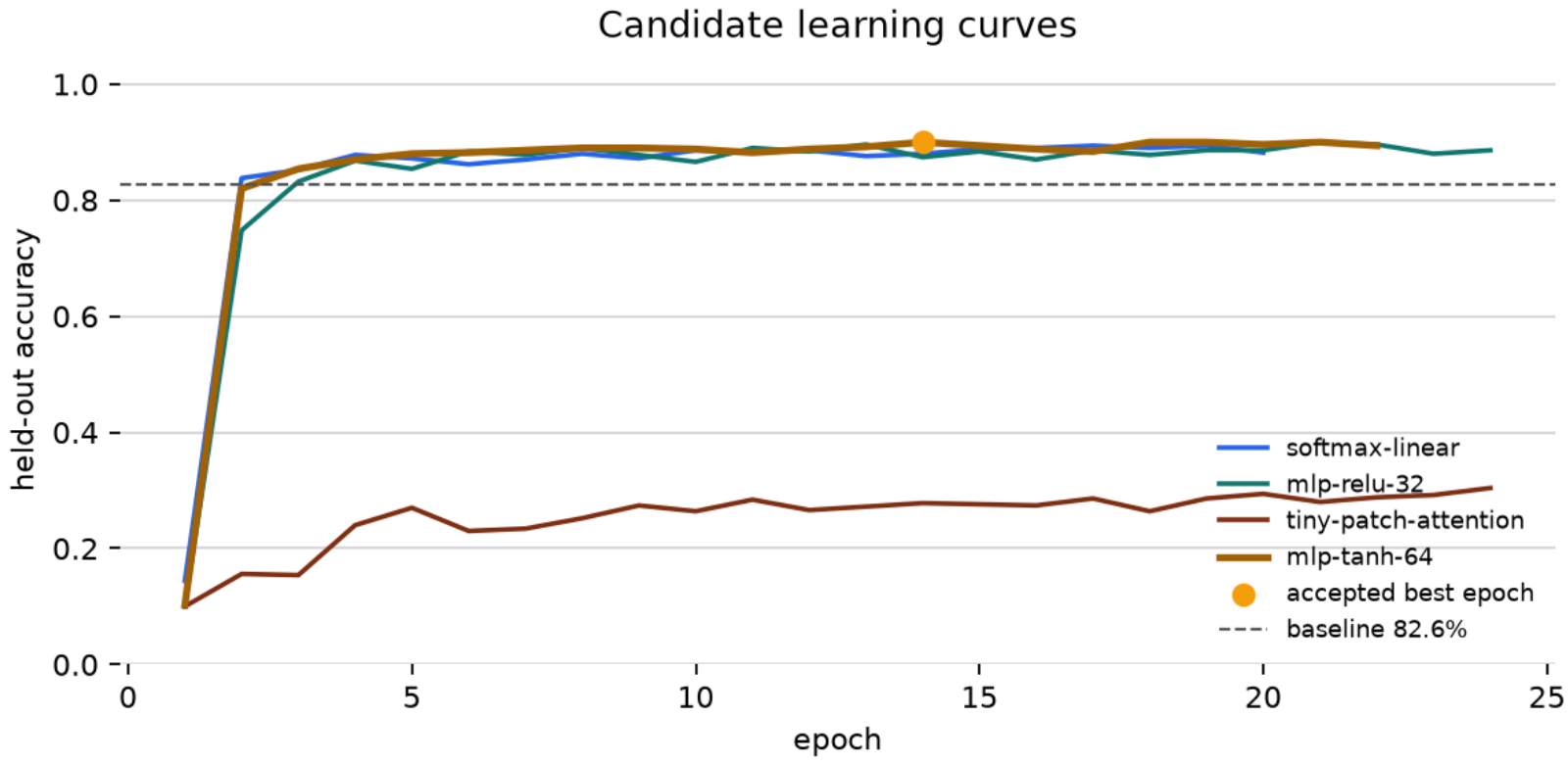


Figure 10: Epoch-level held-out accuracy curves for evaluated candidates from output/data/ml_training_history.csv; the accepted curve is visually emphasized for exp-mlp-tanh-64. Generation method: Epoch-level held-out accuracy lines with accepted best-epoch marker. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics I



Figure 11: Configured-training dynamics for evaluated candidates from output/data/ml_training_diagnostics.json; exp-mlp-tanh-64 is highlighted while best-epoch markers and train-test gaps remain bounded to the local run. Generation method: Final and best-epoch accuracy bars plus train-test gap bars. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics I

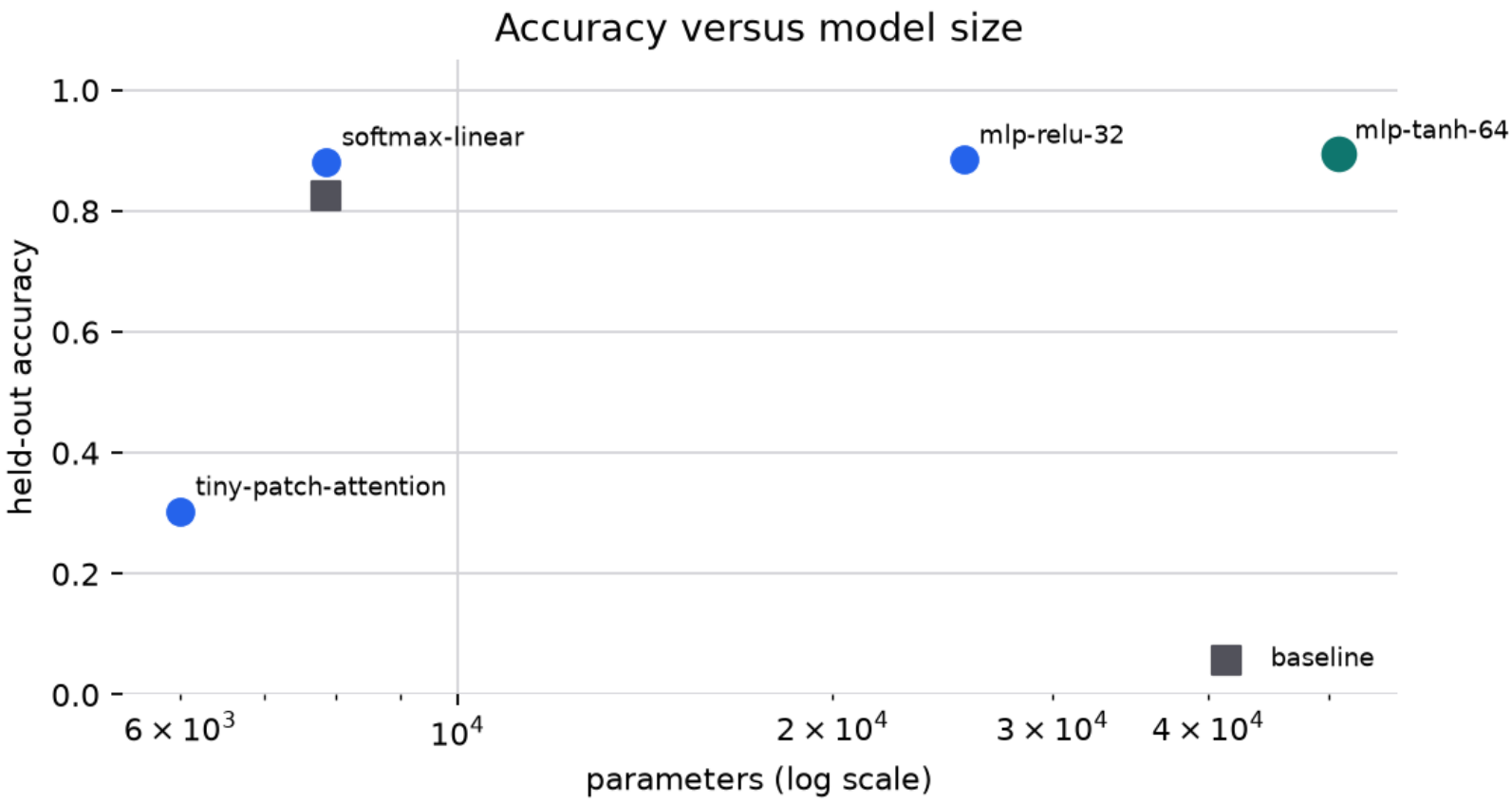


Figure 12: Parameter-count versus held-out accuracy for the baseline and evaluated candidates from output/data/ml_task_results.json; the accepted candidate is highlighted without claiming a general scaling law. Generation method: Log-parameter scatter plot against held-out accuracy. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics I

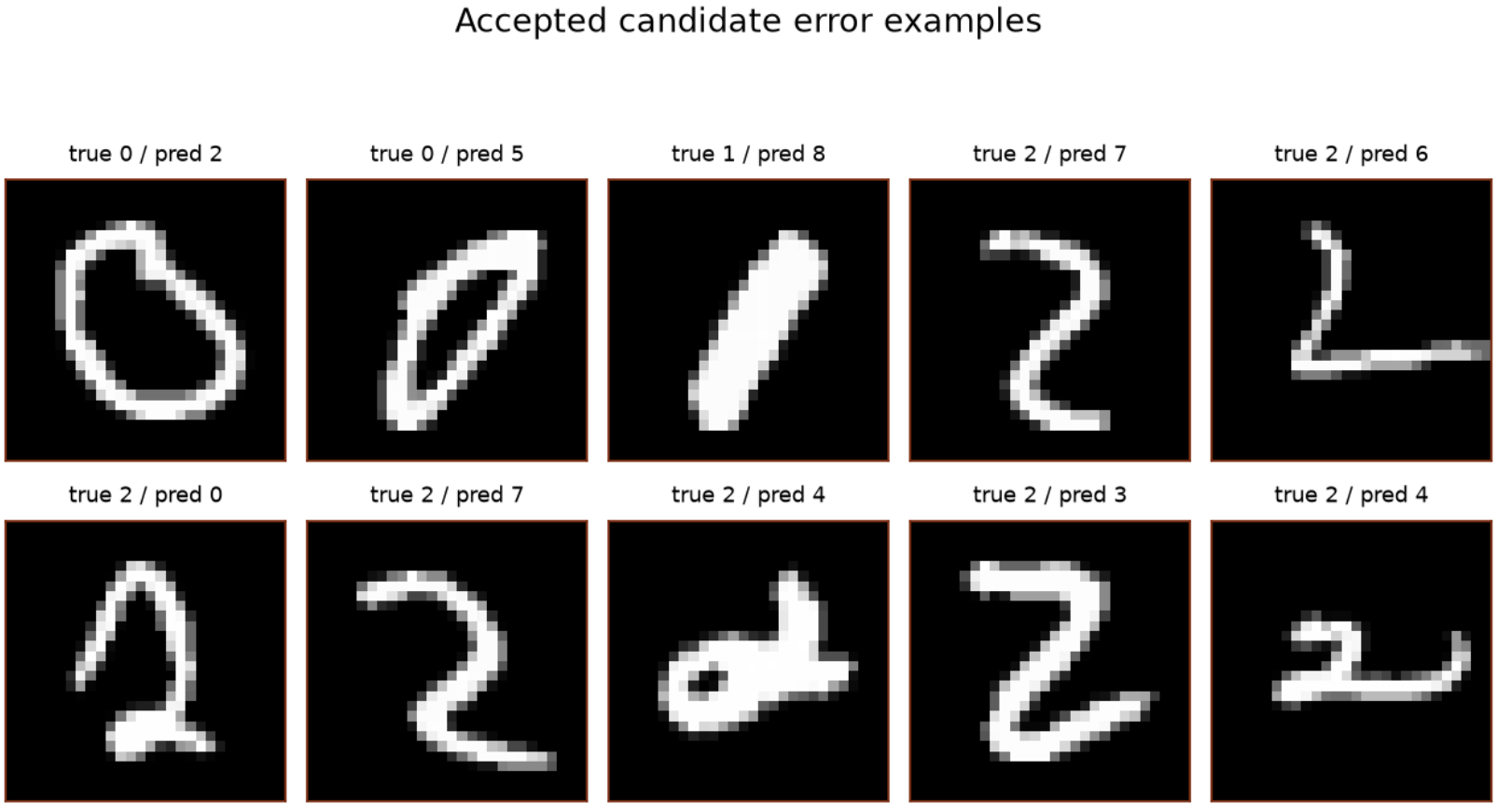


Figure 13: First accepted-candidate error examples for exp-mlp-tanh-64, sourced from output/data/ml_error_examples.json and data/mnist_small.npz; these images support qualitative diagnosis only. Generation method: Deterministic grid of the first accepted-candidate misclassifications. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics I

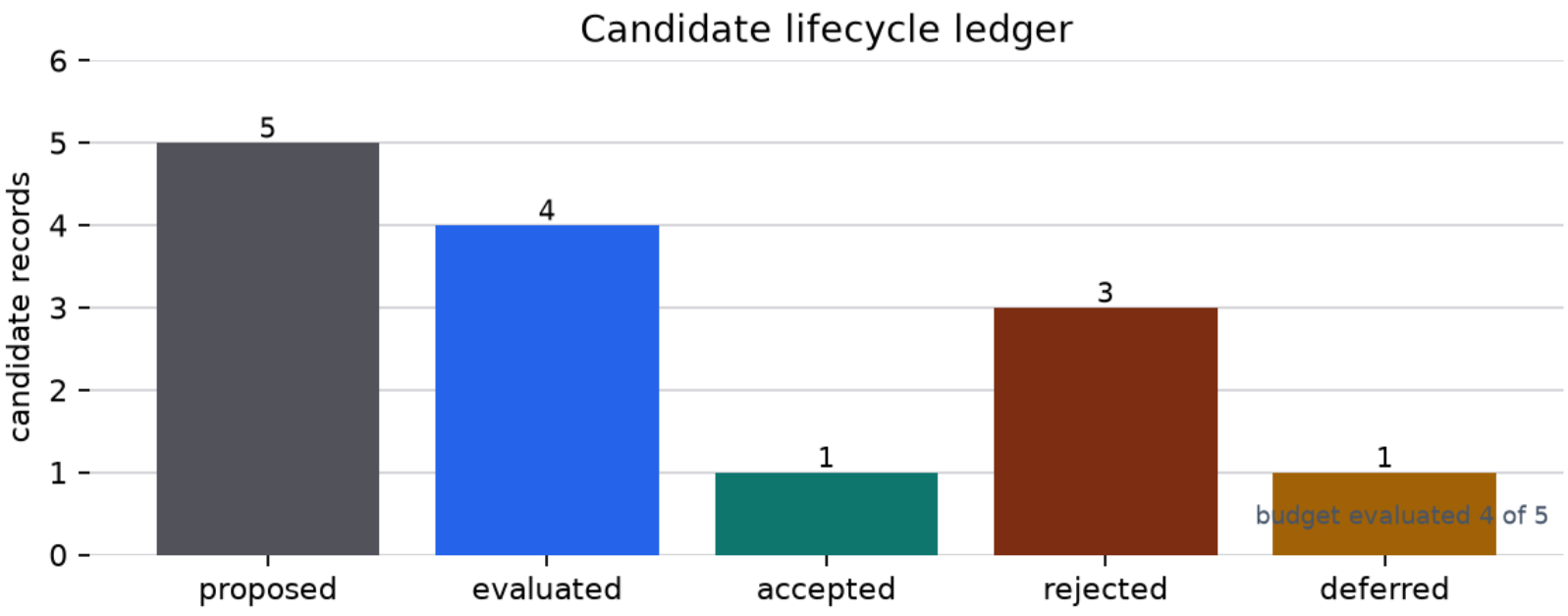


Figure 14: Candidate lifecycle ledger from output/data/ml_candidate_ledger.json: 4 evaluated out of 5 proposed candidates, with deferred proposals kept visible instead of executed automatically. Generation method: Candidate lifecycle status-count bar chart. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Training And Error Diagnostics I

The selected candidate's best held-out epoch is `14`, its final learning rate is `0.144`, its train-loss reduction is `4.161`, and its final train-test accuracy gap is `7.0%`. These values come from the configured training diagnostics rather than from a manually maintained result summary.

Training And Error Diagnostics I

Table 3: Configured-training diagnostics from `output/data/ml_training_diagnostics.json`.

Candidate	Status	Best epoch	Best test accuracy	Final test accuracy	Train-test gap	Loss reduction	Final learning rate
softmax linear	rejected	17	89.4%	88.2%	5.9%	2.917	0.273
mlp relu 32	rejected	21	90.0%	88.6%	8.5%	4.341	0.178
tiny patch attention	rejected	24	30.4%	30.4%	-1.8%	0.165	0.214
mlp tanh 64	accepted	14	90.0%	89.4%	7.0%	4.161	0.144

Diagnostic Analysis

Diagnostic Analysis I

The selected candidate macro F1 is 89.4%, with a held-out accuracy interval of 86.4% to 91.8%. Probability diagnostics report expected calibration error 2.9% and 11 high-confidence error(s). The top non-diagonal confusion pair is 4 -> 9 (4), and the minimum selected-candidate accuracy across deterministic robustness transforms is 80.0%. These values are descriptive diagnostics for the validated local run, not external benchmark claims.

Classification And Error Structure I

The class-level and error-pattern diagnostics are intentionally visual-first: fig. 15 checks confidence calibration, fig. 16 separates precision, recall, and F1 by digit, and fig. 17 ranks the non-diagonal confusions that most shape the selected-candidate error profile. The accompanying tables preserve the same values for audit and downstream comparison.

Classification And Error Structure I

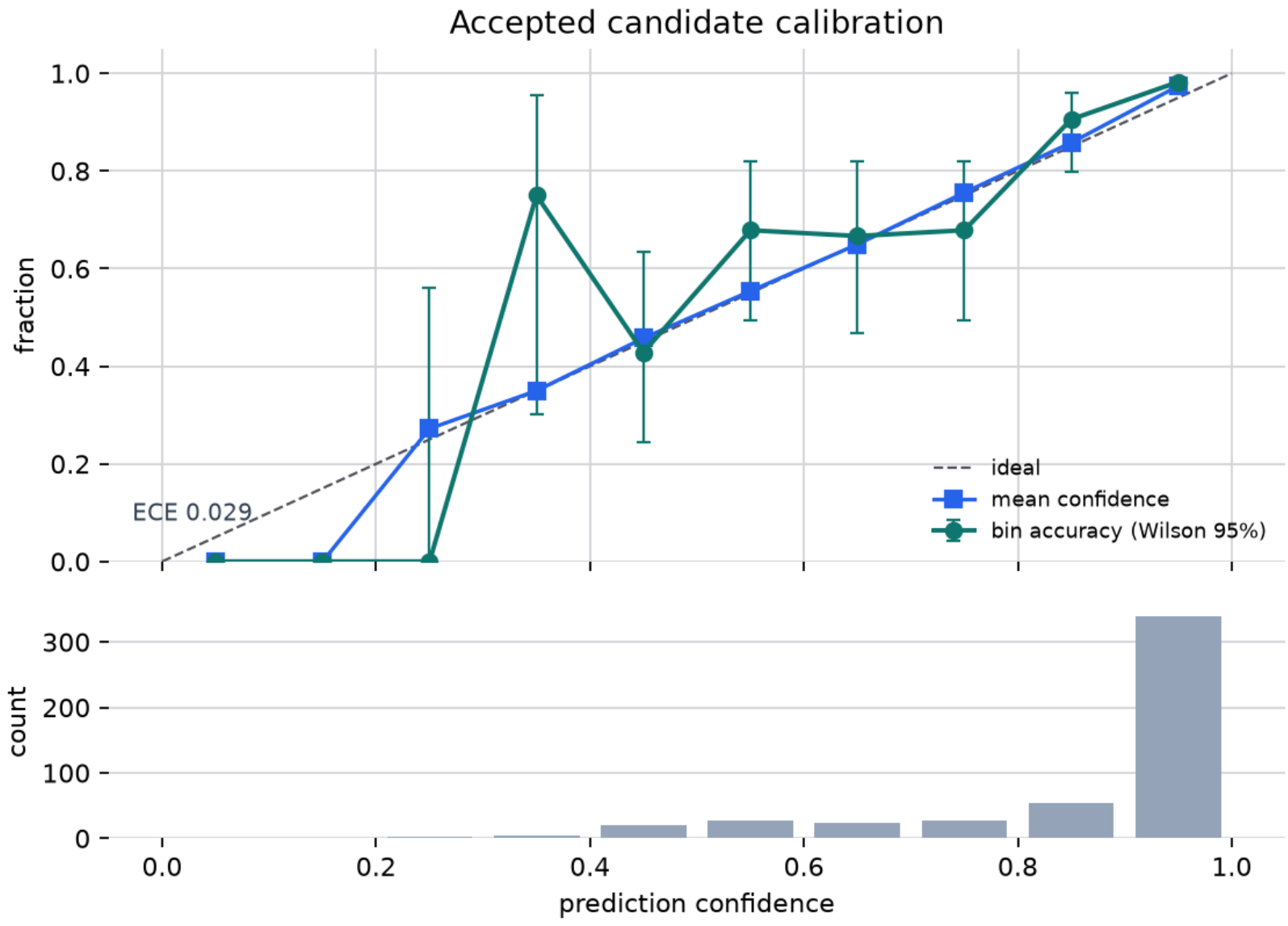


Figure 15: Reliability curve for exp-mlp-tanh-64 from output/data/ml_calibration_report.json; expected calibration error and bin counts summarize the accepted candidate on the fixed local test split. Generation method: Reliability curve with confidence-bin support histogram. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Classification And Error Structure I

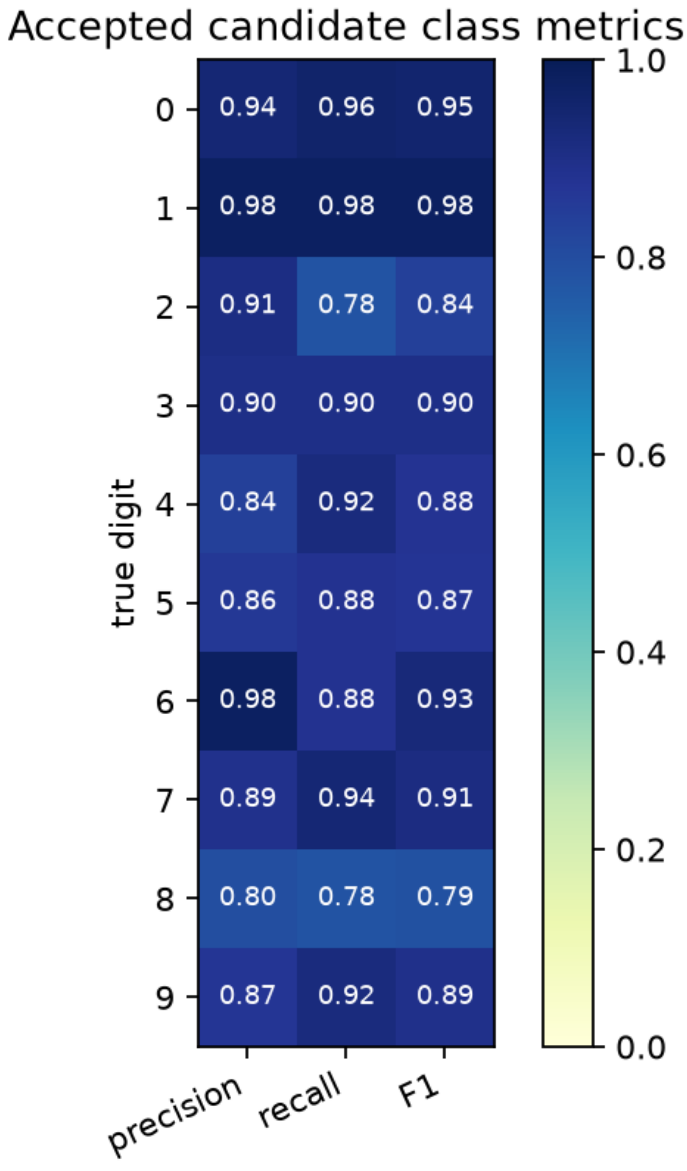


Figure 16: Per-class precision, recall, and F1 for exp-mlp-tanh-64, sourced from output/data/ml_classification_diagnostics.json; metrics are scoped to the local test split. Generation method: Per-class precision, recall, and F1 heatmap. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Classification And Error Structure I

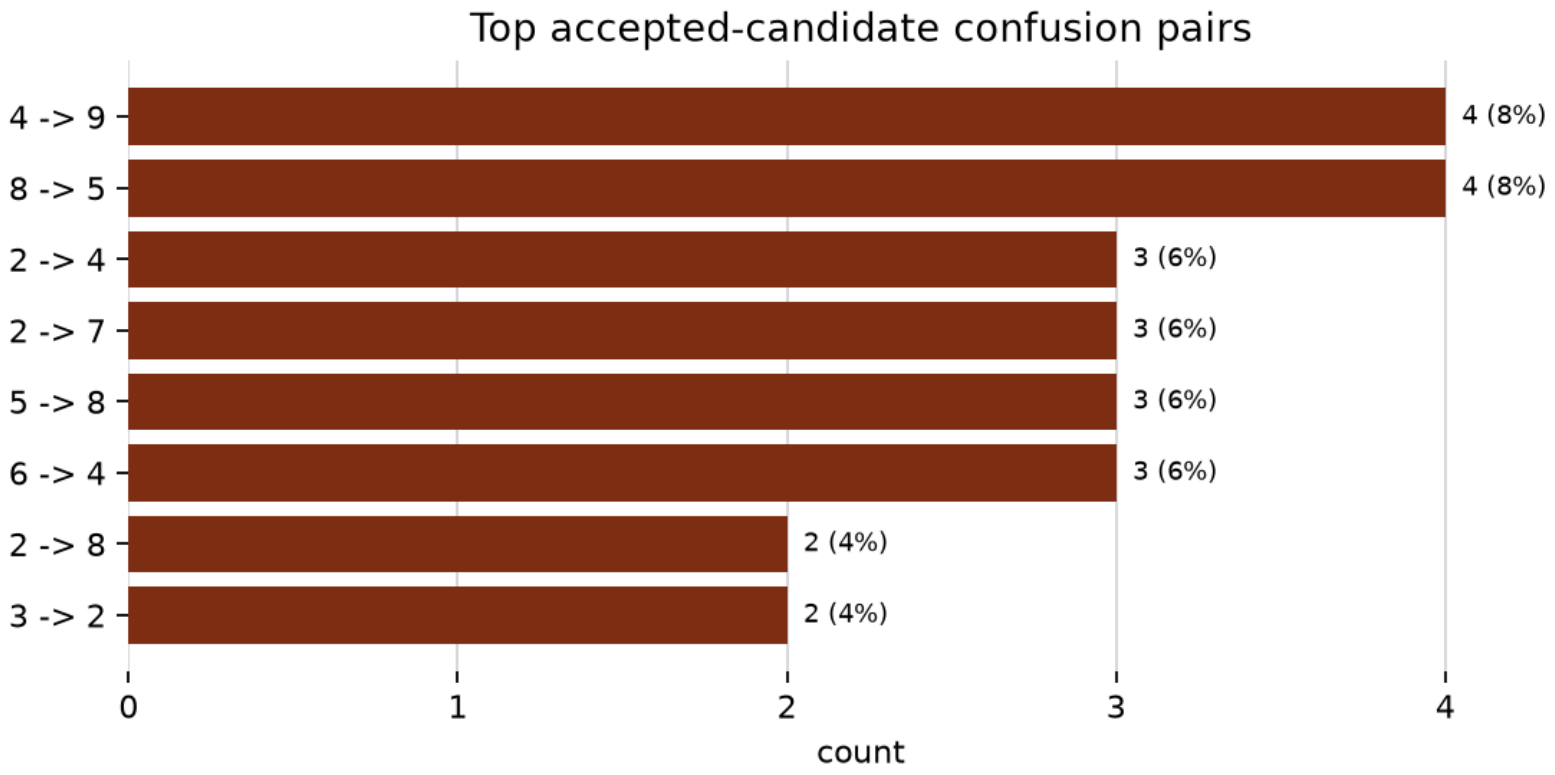


Figure 17: Top non-diagonal confusion pairs for exp-mlp-tanh-64, sourced from output/data/ml_classification_diagnostics.json; the bars highlight which local digit pairs account for accepted-candidate errors. Generation method: Ranked off-diagonal confusion-pair bars with true-class error rates. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Classification And Error Structure I

Table 4: Accepted-candidate class diagnostics from `output/data/ml_classification_diagnostics.json`.

Class	Precision	Recall	F1	Support
0	94.1%	96.0%	95.0%	50
1	98.0%	98.0%	98.0%	50
2	90.7%	78.0%	83.9%	50
3	90.0%	90.0%	90.0%	50
4	83.6%	92.0%	87.6%	50
5	86.3%	88.0%	87.1%	50
6	97.8%	88.0%	92.6%	50
7	88.7%	94.0%	91.3%	50

Classification And Error Structure I

Class	Precision	Recall	F1	Support
8	79.6%	78.0%	78.8%	50
9	86.8%	92.0%	89.3%	50

Classification And Error Structure II

Table 5: Calibration bins from `output/data/ml_calibration_report.json`.

Confidence bin	Count	Accuracy	Mean confidence	Gap
0-0.1	0	0.0%	0.0%	0.0%
0.1-0.2	0	0.0%	0.0%	0.0%
0.2-0.3	3	0.0%	27.2%	27.2%
0.3-0.4	4	75.0%	35.0%	40.0%
0.4-0.5	21	42.9%	45.7%	2.9%
0.5-0.6	28	67.9%	55.4%	12.5%
0.6-0.7	24	66.7%	65.0%	1.7%
0.7-0.8	28	67.9%	75.6%	7.7%

Classification And Error Structure I

Confidence bin	Count	Accuracy	Mean confidence	Gap
0.8-0.9	53	90.6%	85.8%	4.8%
0.9-1	339	98.2%	97.4%	0.8%

Classification And Error Structure II

Table 6: Calibration-bin Wilson intervals from `output/data/ml_calibration_bin_intervals.json`; empty bins are reported explicitly.

Confidence bin	Count	Correct	Accuracy	Wilson 95%	Empty
0-0.1	0	0	0.0%	0.0% to 0.0%	True
0.1-0.2	0	0	0.0%	0.0% to 0.0%	True
0.2-0.3	3	0	0.0%	0.0% to 56.2%	False
0.3-0.4	4	3	75.0%	30.1% to 95.4%	False
0.4-0.5	21	9	42.9%	24.5% to 63.5%	False
0.5-0.6	28	19	67.9%	49.3% to 82.1%	False
0.6-0.7	24	16	66.7%	46.7% to 82.0%	False
0.7-0.8	28	19	67.9%	49.3% to 82.1%	False

Classification And Error Structure I

Confidence bin	Count	Correct	Accuracy	Wilson 95%	Empty
0.8-0.9	53	48	90.6%	79.7% to 95.9%	False
0.9-1	339	333	98.2%	96.2% to 99.2%	False

Classification And Error Structure II

Table 7: Top non-diagonal confusion pairs from `output/data/ml_classification_diagnostics.json`.

Pair	Count	True-class error rate
4 -> 9	4	8.0%
8 -> 5	4	8.0%
2 -> 4	3	6.0%
2 -> 7	3	6.0%
5 -> 8	3	6.0%
6 -> 4	3	6.0%
2 -> 8	2	4.0%
3 -> 2	2	4.0%

Classification And Error Structure I

Pair	Count	True-class error rate
3 -> 8	2	4.0%
5 -> 3	2	4.0%

Uncertainty, Generalization, And Perturbation I

Additional uncertainty and matched-comparison diagnostics report bootstrap accuracy interval 86.4% to 92.0%, bootstrap macro-F1 interval 86.5% to 91.9%, paired net accuracy gain 6.8%, exact McNemar p-value 0.000, mean correct-prediction confidence 90.9%, mean error confidence 63.1%, and 22 low-margin selected-candidate prediction(s). These diagnostics are generated from the fixed local test split and are reported as run evidence, not as population-level certification.

fig. 18 compares train and test behavior, fig. 19 shows deterministic no-retrain perturbation results, fig. 20 summarizes confidence and margin distributions, fig. 21 shows local resampling intervals, and fig. 22 exposes the matched selected-versus-baseline correctness table used by the paired comparison.

Uncertainty, Generalization, And Perturbation I

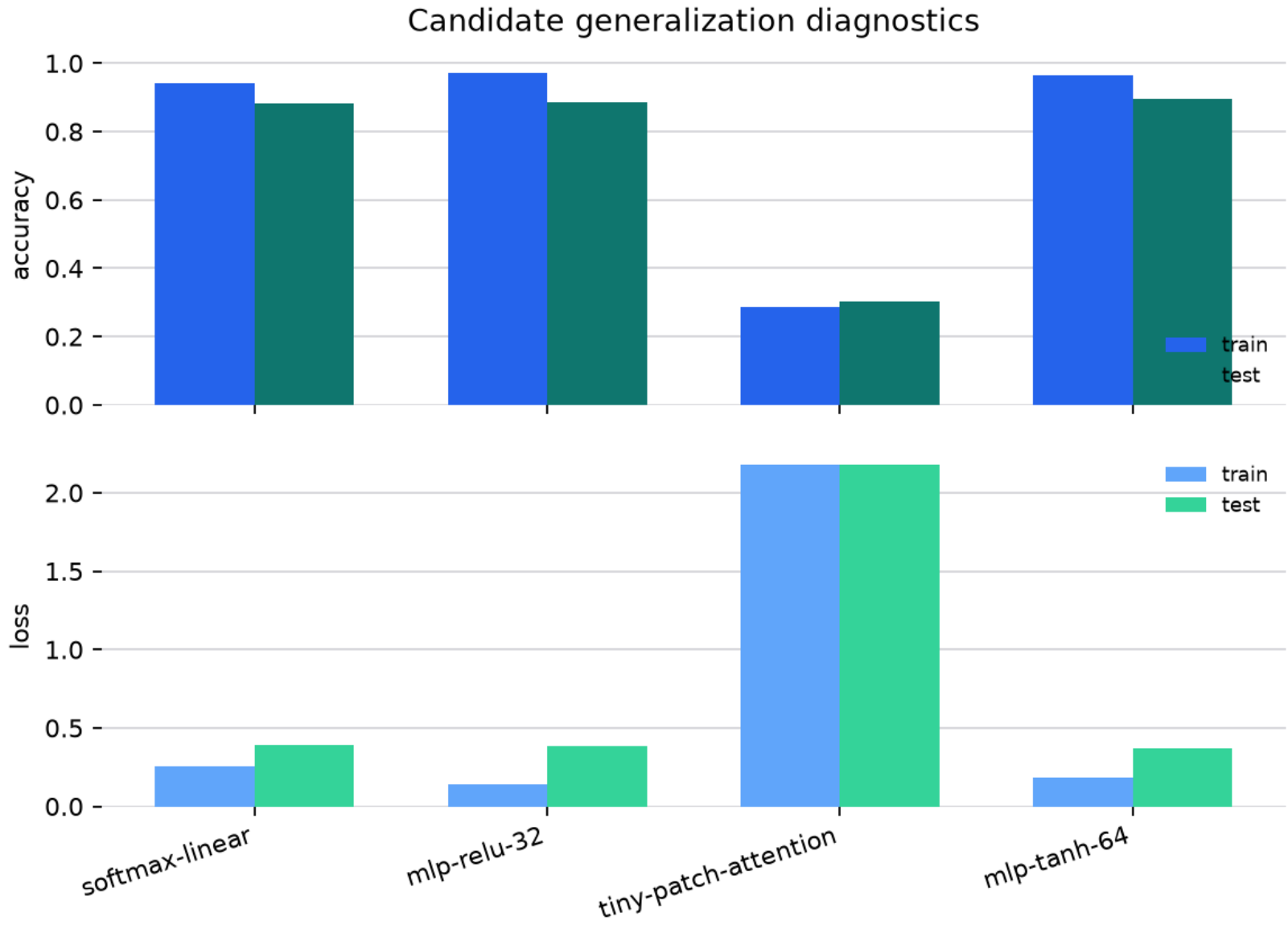


Figure 18: Train/test accuracy and loss for evaluated candidates from output/data/ml_classification_diagnostics.json; the plot exposes local generalization gaps without claiming full-dataset behavior. Generation method: Grouped train/test accuracy and loss bars by evaluated candidate. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Uncertainty, Generalization, And Perturbation I

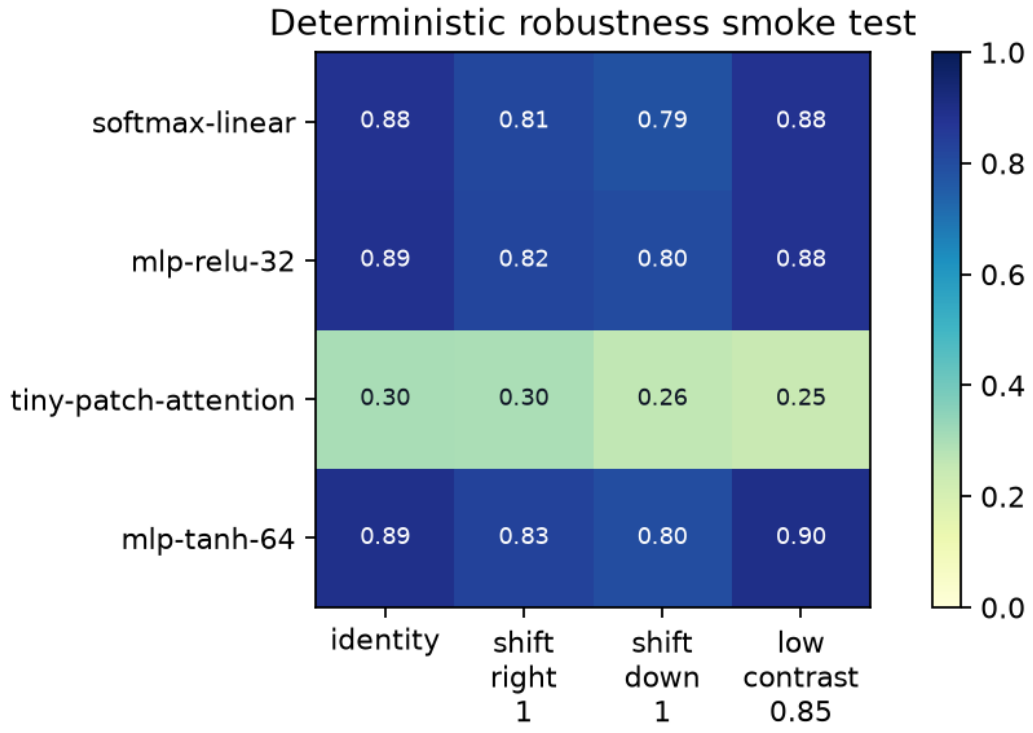


Figure 19: Accuracy for evaluated candidates under identity, one-pixel shifts, and low contrast from output/data/ml_robustness_report.json; the deterministic transforms provide a bounded smoke test only. Generation method: Candidate-by-transform accuracy heatmap for deterministic perturbations. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Uncertainty, Generalization, And Perturbation I

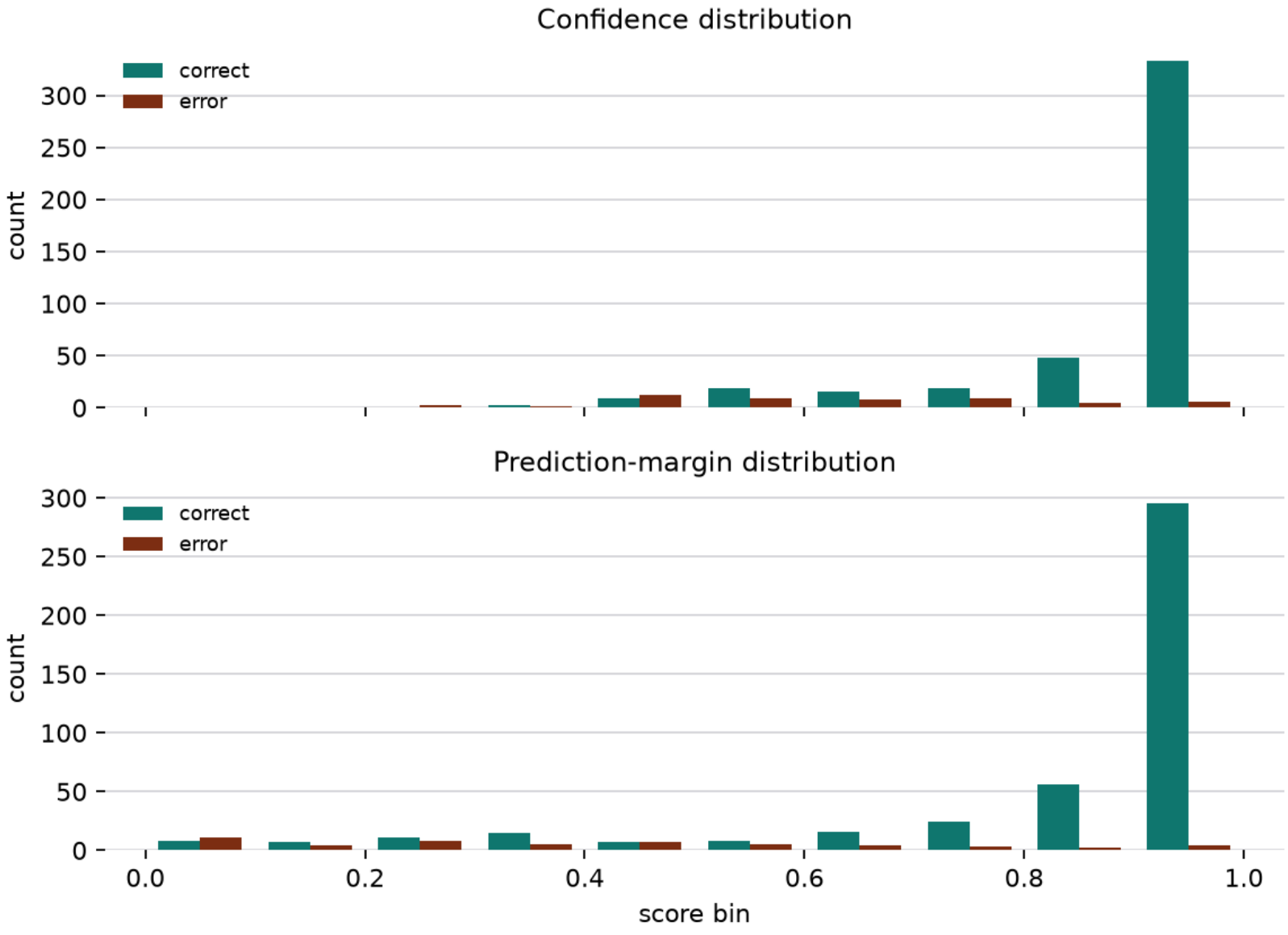


Figure 20: Confidence and prediction-margin histograms for exp-mlp-tanh-64 from output/data/ml_probability_diagnostics.json; the figure separates correct and incorrect local test predictions. Generation method: Confidence and margin histograms split by correctness. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Uncertainty, Generalization, And Perturbation I

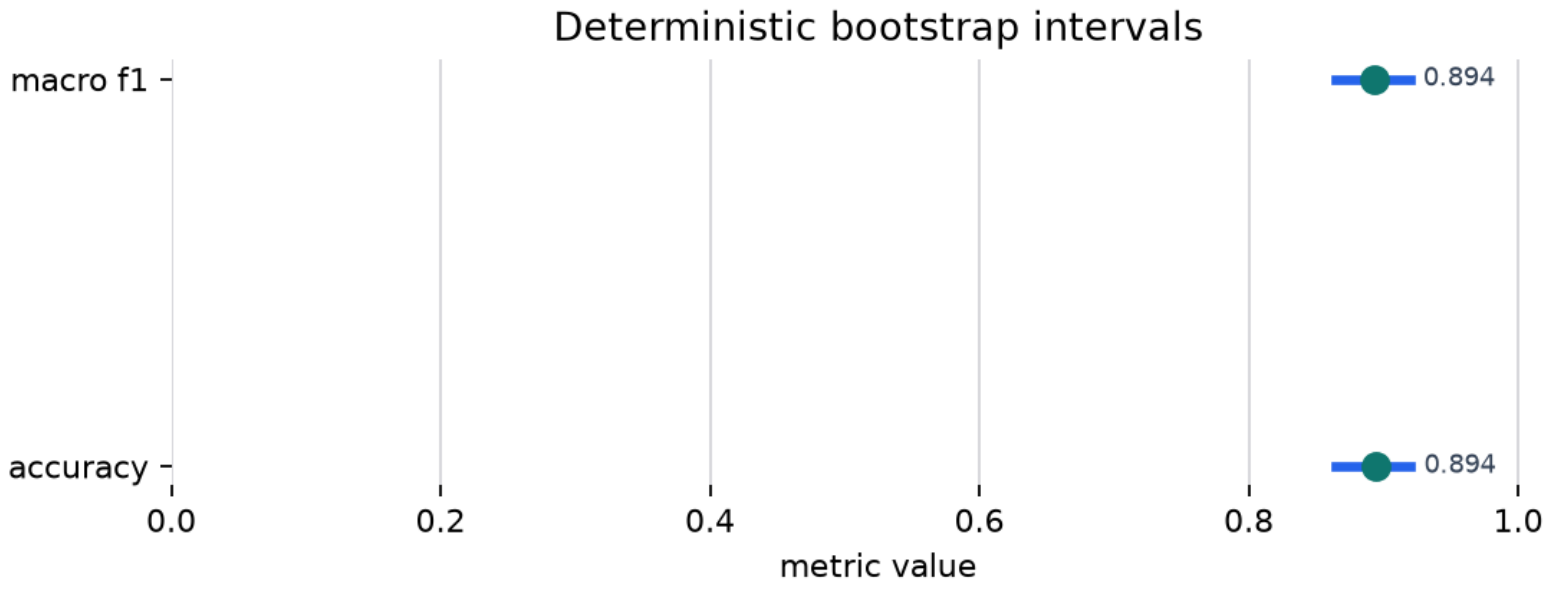
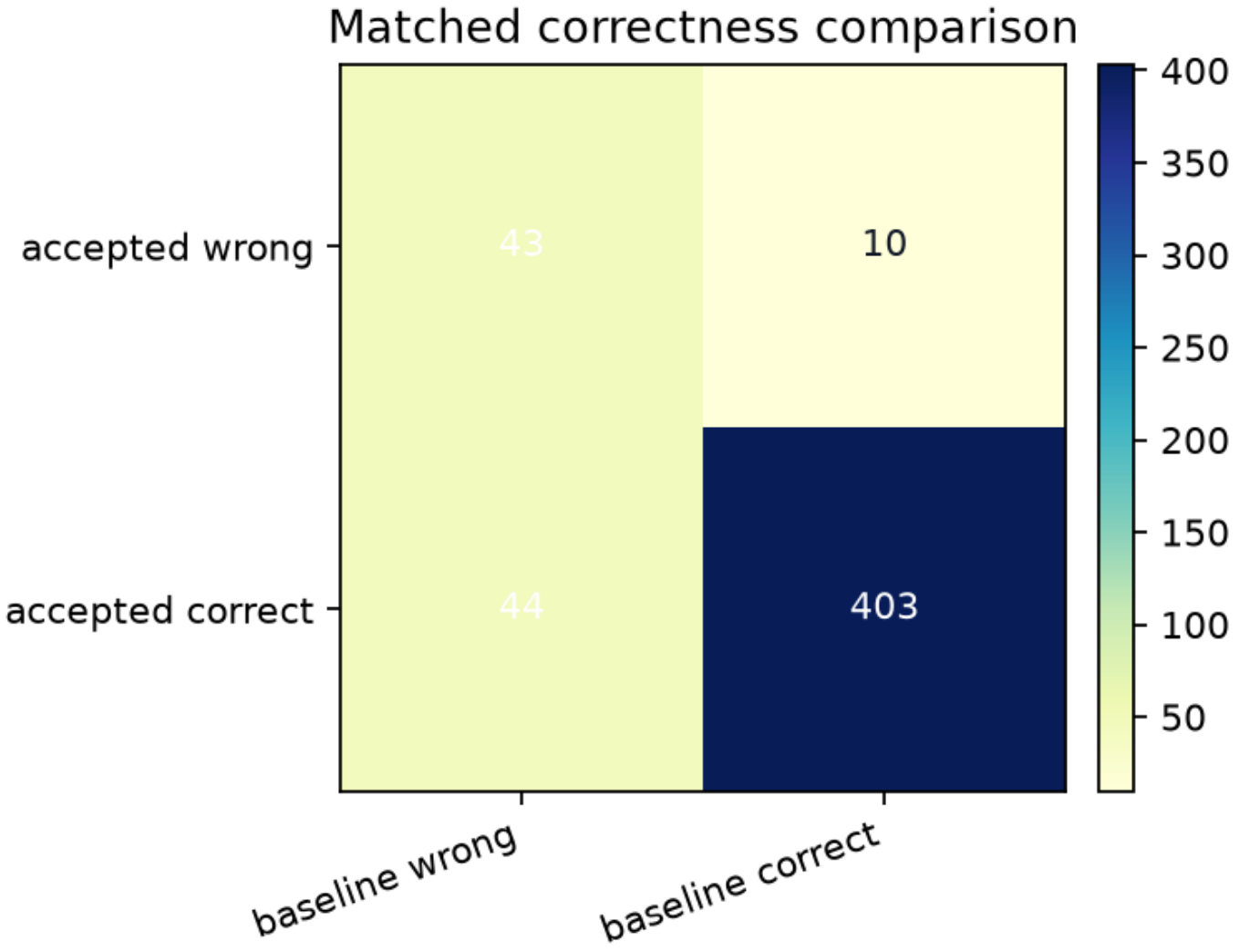


Figure 21: Deterministic percentile-bootstrap intervals for exp-mlp-tanh-64 from output/data/ml_bootstrap_intervals.json; the intervals summarize local sampling variation for accuracy and macro F1. Generation method: Horizontal percentile-bootstrap interval plot. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Uncertainty, Generalization, And Perturbation I



exact McNemar p=0.000

Figure 22: Matched correctness comparison between exp-mlp-tanh-64 and the nearest_centroid_baseline baseline from output/data/ml_paired_comparison.json; discordant cells support the paired test summary. Generation method: Matched accepted-versus-baseline correctness heatmap. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Uncertainty, Generalization, And Perturbation I

Table 8: Deterministic no-retrain robustness smoke test from `output/data/ml_robustness_report.json`.

Candidate	Transform	Accuracy	Samples
softmax linear	identity	88.2%	500
softmax linear	shift_right_1	81.4%	500
softmax linear	shift_down_1	78.6%	500
softmax linear	low_contrast_0_85	88.0%	500
mlp relu 32	identity	88.6%	500
mlp relu 32	shift_right_1	82.4%	500
mlp relu 32	shift_down_1	80.2%	500
mlp relu 32	low_contrast_0_85	88.2%	500

Uncertainty, Generalization, And Perturbation I

Candidate	Transform	Accuracy	Samples
tiny patch attention	identity	30.4%	500
tiny patch attention	shift_right_1	29.8%	500
tiny patch attention	shift_down_1	25.8%	500
tiny patch attention	low_contrast_0_85	24.6%	500
mlp tanh 64	identity	89.4%	500
mlp tanh 64	shift_right_1	83.2%	500
mlp tanh 64	shift_down_1	80.0%	500
mlp tanh 64	low_contrast_0_85	89.8%	500

Uncertainty, Generalization, And Perturbation II

Table 9: Accepted-candidate probability diagnostics from `output/data/ml_probability_diagnostics.json`.

Statistic	Value
Mean confidence	87.9%
Mean correct confidence	90.9%
Mean error confidence	63.1%
Mean margin	80.2%
Mean correct margin	84.9%
Mean error margin	40.3%
Low-margin count	22

Uncertainty, Generalization, And Perturbation III

Table 10: Deterministic percentile-bootstrap intervals from `output/data/ml_bootstrap_intervals.json`.

Metric	Observed	CI low	CI high	Resample mean
accuracy	89.4%	86.4%	92.0%	89.4%
macro F1	89.4%	86.5%	91.9%	89.3%

Uncertainty, Generalization, And Perturbation IV

Table 11: Accepted-candidate versus baseline paired comparison from `output/data/ml_paired_comparison.json`.

Matched comparison statistic	Value
Both correct	403
Accepted only correct	44
Baseline only correct	10
Both wrong	43
Discordant examples	54
Exact McNemar p	0.000
Net accuracy gain	6.8%

Probability Quality And Selective Prediction I

Probability-quality diagnostics report Brier score `0.161`, negative log likelihood `0.361`, top-2 accuracy `95.6%`, and Cohen kappa `0.882` for the selected candidate. At the highest configured confidence threshold, retained coverage is `67.8%` and selective accuracy is `98.2%`.

The selective-prediction view in fig. 23 reports the configured confidence-threshold trade-off: retaining fewer predictions can raise selective accuracy, but the coverage table keeps that trade-off explicit. The candidate probability-quality comparison in fig. 24 keeps accuracy separate from proper-score behavior, so the selected candidate is not treated as automatically best on every diagnostic axis.

Probability Quality And Selective Prediction I

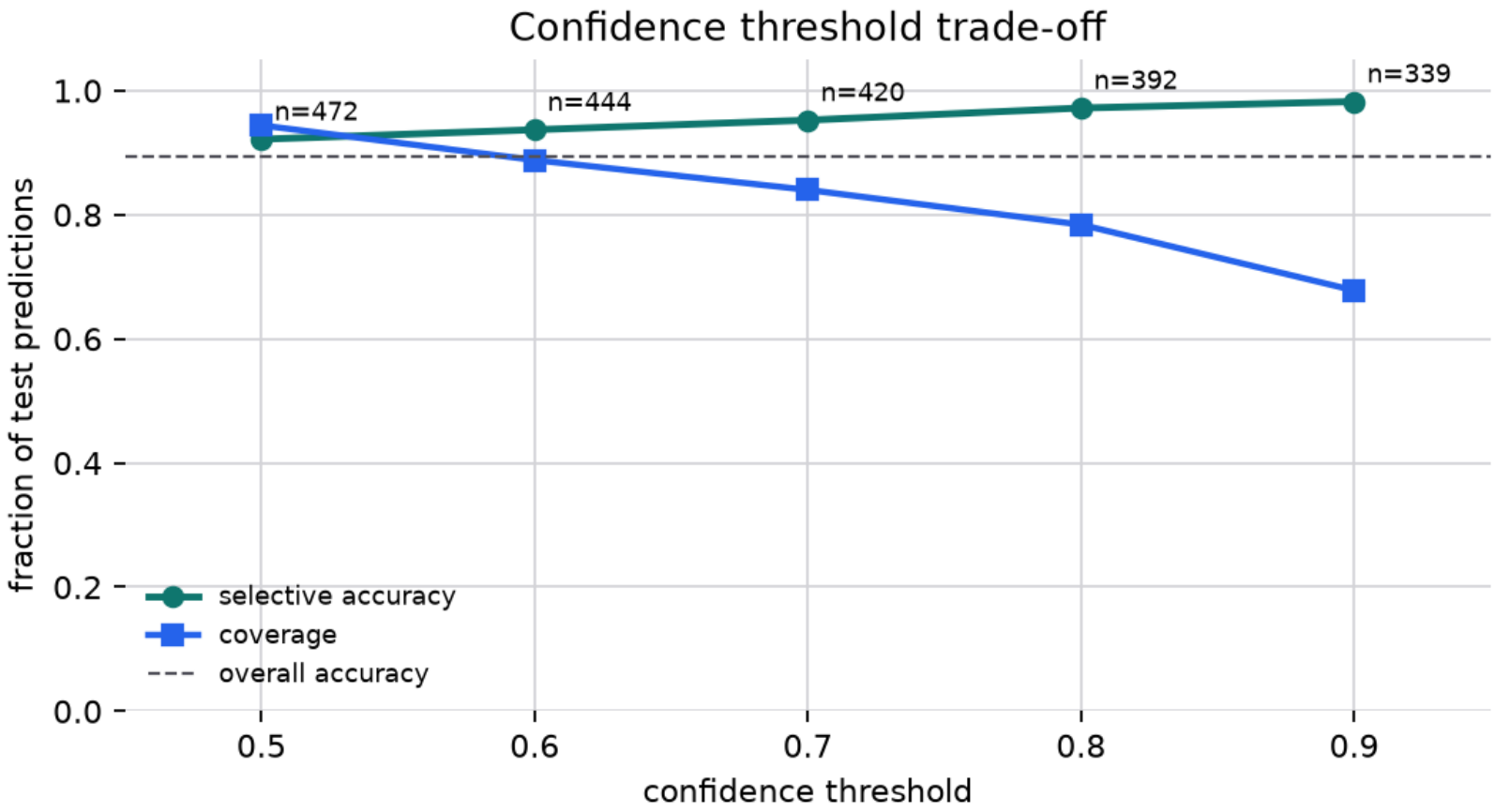


Figure 23: Confidence-threshold trade-off for exp-mlp-tanh-64 from output/data/ml_statistical_summary.json; the plot compares retained coverage, selective accuracy, and the unthresholded accepted-candidate accuracy on the fixed local split. Generation method: Confidence-threshold coverage and selective-accuracy line chart. Registry metadata records the generation method, source artifact, and claim boundary for

Probability Quality And Selective Prediction I

validation.

Probability Quality And Selective Prediction I



Figure 24: Brier score and negative log likelihood for evaluated candidates from output/data/ml_statistical_summary.json; lower values indicate better probability quality within the configured local run, and the accepted candidate is highlighted. Generation method: Brier score and negative-log-likelihood bar comparison. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Probability Quality And Selective Prediction I

Probability Quality And Selective Prediction II

Probability Quality And Selective Prediction I

Table 12: Accepted-candidate statistical summary from `output/data/ml_statistical_summary.json`.

Statistic	Value
Accuracy	89.4%
Balanced accuracy	89.4%
Macro F1	89.4%
Top-2 accuracy	95.6%
Cohen kappa	0.882
Brier score	0.161
Negative log likelihood	0.361
Expected calibration error	2.9%

Probability Quality And Selective Prediction II

Table 13: Selective-accuracy threshold table from `output/data/ml_statistical_summary.json`.

Confidence threshold	Coverage	Selective accuracy	Retained	Errors
50.0%	94.4%	92.2%	472	37
60.0%	88.8%	93.7%	444	28
70.0%	84.0%	95.2%	420	20
80.0%	78.4%	97.2%	392	11
90.0%	67.8%	98.2%	339	6

Probability Quality And Selective Prediction III

Table 14: Candidate probability-quality table from `output/data/ml_statistical_summary.json`.

Candidate	Accuracy	Top-2 accuracy	Brier	NLL	Mean confidence
softmax linear	88.2%	95.8%	0.173	0.390	85.1%
mlp relu 32	88.6%	95.6%	0.164	0.380	89.9%
tiny patch attention	30.4%	46.6%	0.873	2.179	13.1%
mlp tanh 64	89.4%	95.6%	0.161	0.361	87.9%

Candidate Ledger

Candidate Ledger I

Table 15: Candidate lifecycle ledger from `output/data/ml_candidate_ledger.json`.

Candidate	Model	Status	Test accuracy	Parameters
baseline	nearest-centroid	baseline	82.6%	7840
softmax linear	softmax regression	rejected	88.2%	7850
mlp relu 32	MLP	rejected	88.6%	25450
tiny patch attention	tiny patch-attention	rejected	30.4%	5994
mlp tanh 64	MLP	accepted	89.4%	50890
mlp relu 64 deferred	MLP	deferred	N/A	0

Candidate Ledger I

The candidate-selection audit separates the objective ranking from descriptive diagnostics. It records the configured metric, Wilson interval, probability quality, parameter count, and deterministic tie-break context for each evaluated candidate. The diagnostic-boundary table states what each generated surface supports and what it does not support.

Candidate Ledger I

Table 16: Candidate-selection audit from `output/data/ml_candidate_selection_audit.json`; the objective metric ranks candidates, while probability diagnostics and parameter counts audit the chosen tie-break context.

Rank	Candidate	Status	Accuracy	Wilson 95%	Brier	NLL	Parameters
1	mlp tanh 64	accepted	89.4%	86.4% to 91.8%	0.161	0.361	50890
2	mlp relu 32	rejected	88.6%	85.5% to 91.1%	0.164	0.380	25450
3	softmax linear	rejected	88.2%	85.1% to 90.7%	0.173	0.390	7850
4	tiny patch attention	rejected	30.4%	26.5% to 34.6%	0.873	2.179	5994

Candidate Ledger I

Table 17: Diagnostic claim-boundary table from `output/data/ml_diagnostic_boundary.json`.

Surface	Source	Method	Supports	Does not support
objective selection	task results	rank evaluated candidates by configured held-out metric and deterministic tie...	accepted-candidate selection within this fixed local task	full MNIST state-of-the-art, external benchmark leadership, or universal mode...
descriptive diagnostics	class diagnostics	per-class metrics, calibration, probability quality, and paired comparison	local error analysis and uncertainty description	population-level certification or deployment readiness
robustness smoke test	robustness report	deterministic no-retrain transforms applied to the fixed test split	small perturbation smoke-test evidence	adversarial robustness or distribution-shift robustness
artifact integrity	integrity attestation	local SHA-256 checks over declared inputs and generated artifacts	local artifact integrity evidence for the run	external signing, production SLSA compliance, or runtime intrusion detection
review governance	review decisions	deferred generated review decisions with human review required	readiness for human review	machine self-approval or publication acceptance

Readiness And Review Artifacts

Readiness And Review Artifacts I

The broader AutoResearch run writes the reproducibility, benchmark, review, and manuscript-hydration surfaces summarized below. The schema manifest records 31 schema-versioned governance payload(s), and the local research-object manifest records 84 observed artifact record(s) with checksums and approval state `false`. The phase ledger records 3 settlement pass(es), while the figure-quality report covers 25 registered figure(s) with validity `false`.

Readiness And Review Artifacts I

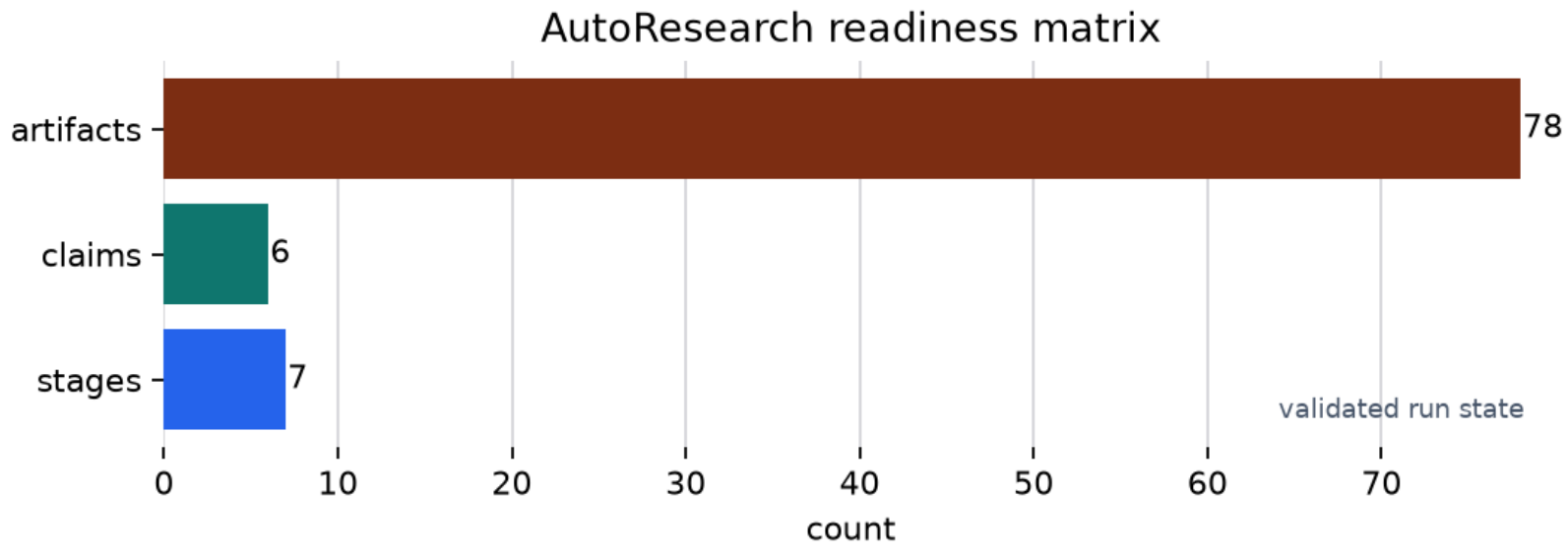


Figure 25: Validated AutoResearch run with 7 stages, 6 supported claims, and 78 required artifacts from output/data/autoresearch_loop.json; the count summarizes readiness artifacts, not human approval. Generation method: Horizontal count summary from final loop metrics. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Readiness And Review Artifacts I

File-backed AutoResearch closure



Figure 26: File-backed AutoResearch closure from program through review, with 6 supported claims and readiness passed; review remains a deferred human decision and the provenance path remains inspectable. Generation method: File-backed process-flow diagram from final loop state. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Readiness And Review Artifacts I

Readiness And Review Artifacts II

Readiness And Review Artifacts I

Table 18: Reviewed upstream analysis and evidence paths from `output/reports/artifact_manifest.json`; unreviewed paths are omitted, and that terminal manifest remains the complete byte authority.

Artifact	Role
output/data/autoresearch_claims.json	Loop artifact
output/data/autoresearch_evidence_overview.json	Evidence registry
output/data/autoresearch_integrity_attestation.json	Security evidence
output/data/autoresearch_inventory_export.json	Security evidence
output/data/autoresearch_loop.json	Loop artifact
output/data/autoresearch_phase_ledger.json	Run or candidate ledger
output/data/autoresearch_plan.json	Loop artifact
output/data/autoresearch_review_packet.json	Review packet

Readiness And Review Artifacts I

Artifact	Role
output/data/autoresearch_schema_manifest.json	Loop artifact
output/data/autoresearch_security_profile.json	Security evidence
output/data/autoresearch_stage_matrix.csv	Loop artifact
output/data/autoresearch_supply_chain_inventory.json	Security evidence
output/data/autoresearch_threat_model.json	Security evidence
output/data/benchmark_boundary.json	Benchmark grading
output/data/benchmark_scores.json	Benchmark grading
output/data/figure_quality_report.json	Loop artifact

Readiness And Review Artifacts I

Artifact	Role
output/data/figure_style.json	Loop artifact
output/data/idea_ledger.json	Run or candidate ledger
output/data/ml_bootstrap_intervals.json	Loop artifact
output/data/ml_calibration_bin_intervals.json	Loop artifact
output/data/ml_calibration_report.json	Loop artifact
output/data/ml_candidate_intervals.json	Loop artifact
output/data/ml_candidate_ledger.json	Run or candidate ledger
output/data/ml_candidate_rank_stability.json	Loop artifact

Readiness And Review Artifacts I

Artifact	Role
output/data/ml_candidate_selection_audit.json	Loop artifact
output/data/ml_class_balance.json	Loop artifact
output/data/ml_classification_diagnostics.json	Loop artifact
output/data/ml_confusion_matrix.csv	Loop artifact
output/data/ml_diagnostic_boundary.json	Loop artifact
output/data/ml_error_examples.json	Loop artifact
output/data/ml_paired_comparison.json	Loop artifact
output/data/ml_prediction_records.json	Loop artifact

Readiness And Review Artifacts I

Artifact	Role
output/data/ml_probability_diagnostics.json	Loop artifact
output/data/ml_robustness_report.json	Loop artifact
output/data/ml_statistical_summary.json	Loop artifact
output/data/ml_task_results.json	Loop artifact
output/data/ml_training_diagnostics.json	Loop artifact
output/data/ml_training_history.csv	Loop artifact
output/data/mnist_task_config.json	Loop artifact
output/data/research_object_manifest.json	Loop artifact

Readiness And Review Artifacts I

Artifact	Role
output/data/research_program.json	Loop artifact
output/data/review_decisions.json	Review packet
output/data/run_ledger.json	Run or candidate ledger
output/figures/autoresearch_candidate_lifecycle.png	Generated figure
output/figures/autoresearch_closure_flow.png	Generated figure
output/figures/autoresearch_integrity_chain.png	Generated figure
output/figures/autoresearch_security_control_matrix.png	Generated figure
output/figures/autoresearch_stage_matrix.png	Generated figure

Readiness And Review Artifacts I

Artifact	Role
output/figures/figure_registry.json	Loop artifact
output/figures/ml_bootstrap_intervals.png	Generated figure
output/figures/ml_calibration_reliability.png	Generated figure
output/figures/ml_candidate_rank_stability.png	Generated figure
output/figures/ml_candidate_scores.png	Generated figure
output/figures/ml_classification_metrics_heatmap.png	Generated figure
output/figures/ml_complexity_accuracy.png	Generated figure
output/figures/ml_confusion_matrix.png	Generated figure

Readiness And Review Artifacts I

Artifact	Role
output/figures/ml_confusion_pairs.png	Generated figure
output/figures/ml_generalization_gap.png	Generated figure
output/figures/ml_learning_curves.png	Generated figure
output/figures/ml_paired_correctness.png	Generated figure
output/figures/ml_per_class_accuracy.png	Generated figure
output/figures/ml_probability_margin_distribution.png	Generated figure
output/figures/ml_probability_quality.png	Generated figure
output/figures/ml_robustness_matrix.png	Generated figure

Readiness And Review Artifacts I

Artifact	Role
output/figures/ml_selective_accuracy.png	Generated figure
output/figures/ml_training_dynamics.png	Generated figure
output/figures/mnist_class_balance.png	Generated figure
output/figures/mnist_error_examples.png	Generated figure
output/figures/mnist_subset_contact_sheet.png	Generated figure
output/reports/autoresearch_evidence_overview.md	Evidence registry
output/reports/autoresearch_loop.json	Loop artifact
output/reports/autoresearch_loop.md	Loop artifact

Readiness And Review Artifacts I

Artifact	Role
output/reports/autoresearch_review_packet.md	Review packet
output/reports/autoresearch_security_review.md	Review packet
output/reports/autoresearch_summary.md	Loop artifact
output/reports/benchmark_readiness_smoke.json	Benchmark grading
output/reports/ml_benchmark_score.json	Benchmark grading
output/reports/ml_experiment_report.md	Loop artifact
output/reports/test_results.json	Loop artifact
output/reports/test_results.md	Loop artifact

Readiness And Review Artifacts II

Table 19: Review-gate decisions from `output/data/review_decisions.json`.

Gate	Required	Decision	Rationale
proposal_review	True	deferred	Decision is read from human_review.yaml when present; generated readiness is not approval.
evidence_review	True	deferred	Decision is read from human_review.yaml when present; generated readiness is not approval.

Readiness And Review Artifacts III

Table 20: Benchmark grading table from `output/data/benchmark_scores.json`.

Benchmark task	Status	Score	Grading output
readiness-smoke	graded	1	`output/reports/benchmark_readiness_smoke.json`
ml-loop-score	graded	1	`output/reports/ml_benchmark_score.json`

Readiness And Review Artifacts IV

Table 21: Deterministic phase ledger from `output/data/autoresearch_phase_ledger.json`; settlement order is not an autonomy claim.

Phase	Order	Group	Observed artifacts	Description
intrinsic readiness	1	readiness	3	validate configured project-intrinsic contracts
core artifacts	2	loop	0	write plan, stage matrix, and provisional loop outputs
evidence registry	3	evidence	2	write local evidence registry
ml task	4	ml	54	run fixed-seed bounded candidate evaluation
method contract	5	governance	3	write program, idea, run, review, and benchmark ledgers
provisional payloads	6	settlement	12	refresh loop payloads before extrinsic validation
security artifacts	7	security	10	write local security and integrity evidence
final visuals	8	figures	54	write final registry-backed figures

Readiness And Review Artifacts I

Phase	Order	Group	Observed artifacts	Description
manuscript hydration	9	manuscript	9	write variables, provenance, and figure blocks
readiness manifest	10	settlement	12	refresh checksum manifest before extrinsic validation
schema manifest	11	schema	1	write generated JSON schema-version manifest
research object manifest	12	packaging	1	write local research-object manifest
extrinsic readiness	13	readiness	3	validate generated artifacts and extrinsic contracts
final schema manifest	14	schema	1	refresh schema manifest after final payload updates
final research object manifest	15	packaging	1	refresh local research-object manifest
artifact manifest	16	settlement	12	write final artifact checksum manifest

Readiness And Review Artifacts II

Table 22: Figure-quality checks from `output/data/figure_quality_report.json`; 25 registered figure(s) were checked.

Figure	Pixels	Variance	Source exists	Nonblank
`fig:autoresearch_candidate_lifecycle`	1184x480	0.083	True	True
`fig:autoresearch_closure_flow`	1664x448	0.016	True	True
`fig:autoresearch_integrity_chain`	1440x736	0.041	True	True
`fig:autoresearch_security_control_matrix`	1472x736	0.014	True	True
`fig:autoresearch_stage_matrix`	1120x416	0.080	True	True
`fig:ml_bootstrap_intervals`	1152x448	0.009	True	True
`fig:ml_calibration_reliability`	1152x832	0.016	True	True
`fig:ml_candidate_rank_stability`	1408x608	0.053	True	True

Readiness And Review Artifacts I

Figure	Pixels	Variance	Source exists	Nonblank
fig:ml_candidate_scores	1376×688	0.014	True	True
fig:ml_classification_metrics_heatmap	928×832	0.085	True	True
fig:ml_complexity_accuracy	1120×608	0.012	True	True
fig:ml_confusion_matrix	896×768	0.045	True	True
fig:ml_confusion_pairs	1152×576	0.129	True	True
fig:ml_generalization_gap	1184×864	0.059	True	True
fig:ml_learning_curves	1216×608	0.018	True	True
fig:ml_paired_correctness	768×672	0.074	True	True

Readiness And Review Artifacts I

Figure	Pixels	Variance	Source exists	Nonblank
fig:ml_per_class_accuracy	1152×512	0.099	True	True
fig:ml_probability_margin_distribution	1184×864	0.024	True	True
fig:ml_probability_quality	1344×608	0.046	True	True
fig:ml_robustness_matrix	1280×608	0.078	True	True
fig:ml_selective_accuracy	1088×608	0.017	True	True
fig:ml_training_dynamics	1408×608	0.072	True	True
fig:mnist_class_balance	1216×544	0.071	True	True
fig:mnist_error_examples	1280×736	0.234	True	True

Readiness And Review Artifacts I

Figure	Pixels	Variance	Source exists	Nonblank
fig:mnist_subset_contact_sheet	1216x544	0.230	True	True

Security Readiness And Integrity Evidence

Security Readiness And Integrity Evidence I

The local security profile reports attestation status `passed` after checking 80 file record(s), with 0 missing record(s) and 0 checksum mismatch(es). The inventory contains 14 input record(s) and 69 generated-artifact record(s). The integrity-chain figure is fig. 27. These values support local artifact-integrity claims only; they do not claim external signing, production SLSA compliance, or runtime security monitoring.

Security Readiness And Integrity Evidence I

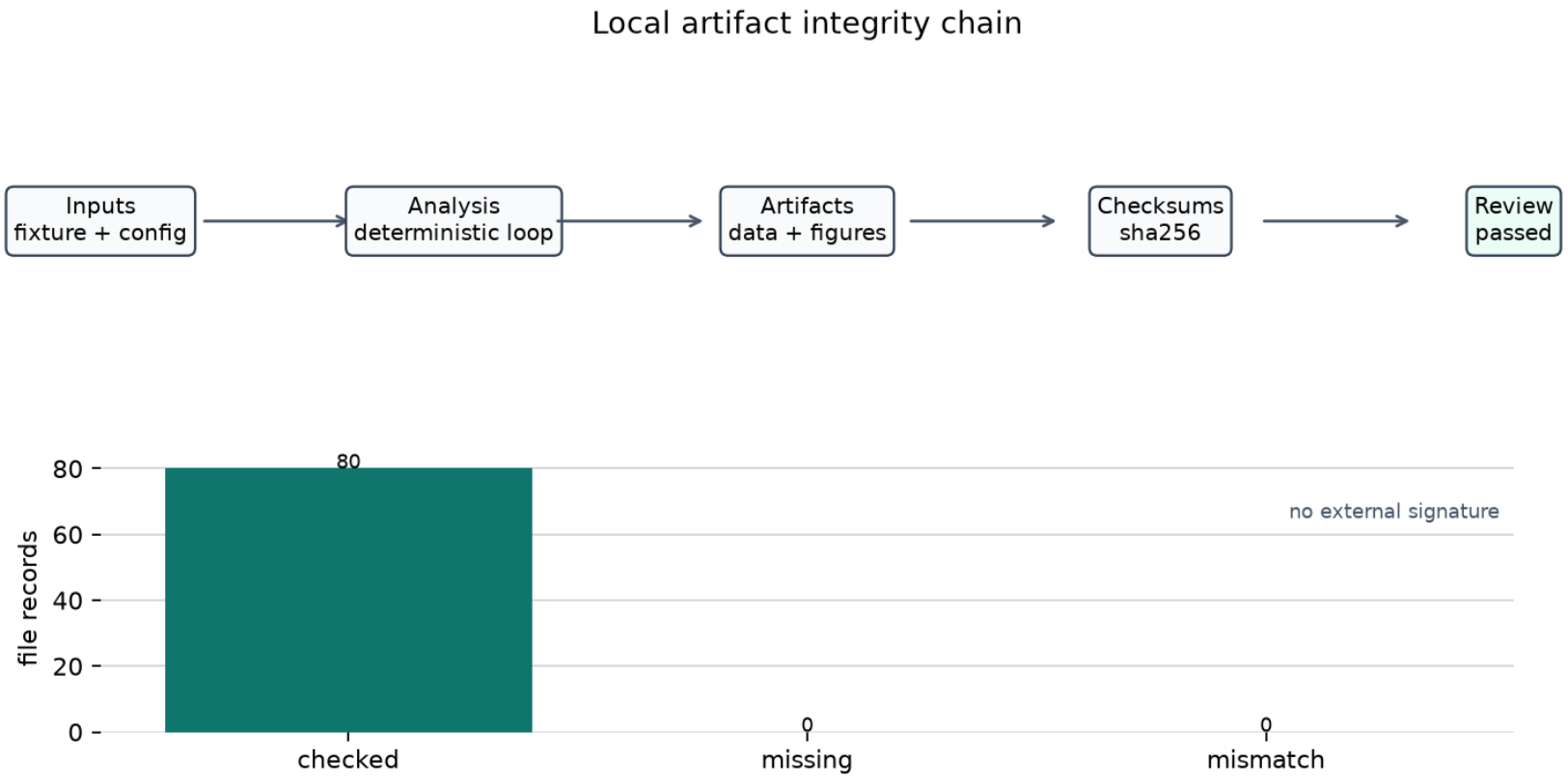


Figure 27: Local integrity chain from output/data/autoresearch_integrity_attestation.json; checksums summarize the observed run artifacts and remain unsigned local evidence. Generation method: Local checksum attestation chain with checked, missing, and mismatch counts. Registry metadata records the generation method, source artifact, and claim boundary for validation.

Security Readiness And Integrity Evidence I

Table 23: Integrity-attestation summary from `output/data/autoresearch_integrity_attestation.json`.

Integrity field	Value
status	passed
algorithm	sha256
checked files	80
missing files	0
checksum mismatches	0
external signature	False

Manuscript Hydration Provenance

Manuscript Hydration Provenance I

The final run supports 6 manuscript-facing claim(s) and checks 78 required artifact(s). The rendered manuscript uses injected variables from generated data payloads, so the abstract and results track the latest analysis run rather than hard-coded counts. The final readiness status is `passed`; generated review decisions are recorded as deferred for human review rather than as self-approval.

Manuscript Hydration Provenance II

Manuscript Hydration Provenance I

Table 24: Variable provenance summary from `output/data/manuscript_variable_provenance.json`.

Source artifact	Injected variables or fragments
output/data/autoresearch_integrity_attestation.json	5
output/data/autoresearch_loop.json	57
output/data/autoresearch_phase_ledger.json	2
output/data/autoresearch_schema_manifest.json	1
output/data/autoresearch_security_profile.json	6
output/data/autoresearch_supply_chain_inventory.json	3
output/data/autoresearch_threat_model.json	4
output/data/benchmark_scores.json	1

Manuscript Hydration Provenance I

Source artifact	Injected variables or fragments
output/data/figure_quality_report.json	3
output/data/manuscript_variable_provenance.json	1
output/data/ml_bootstrap_intervals.json	3
output/data/ml_calibration_bin_intervals.json	1
output/data/ml_calibration_report.json	3
output/data/ml_candidate_intervals.json	1
output/data/ml_candidate_ledger.json	1
output/data/ml_candidate_rank_stability.json	3

Manuscript Hydration Provenance I

Source artifact	Injected variables or fragments
output/data/ml_candidate_selection_audit.json	1
output/data/ml_class_balance.json	3
output/data/ml_classification_diagnostics.json	5
output/data/ml_diagnostic_boundary.json	1
output/data/ml_paired_comparison.json	3
output/data/ml_probability_diagnostics.json	4
output/data/ml_robustness_report.json	2
output/data/ml_statistical_summary.json	9

Manuscript Hydration Provenance I

Source artifact	Injected variables or fragments
output/data/ml_task_results.json	39
output/data/ml_training_diagnostics.json	7
output/data/research_object_manifest.json	2
output/data/review_decisions.json	1
output/figures/figure_registry.json	51
output/reports/artifact_manifest.json	1